Arista Access Node BAS-ACC-62 Critical Failure SOP: Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding)

Node: BAS-ACC-62 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_7563

Description

The critical failure domain of routing protocol flap on Arista Access node BAS-ACC-62, located at Basavanagudi POP, is primarily attributed to the intricacies of the BGP, OSPF, and ISIS routing protocols. In this specific scenario, the failure is characterized by a BGP route churn, OSPF adjacency loss, and ISIS LSP flooding. This occurrence is likely precipitated by the convergence of complex routing protocol interactions, which may be induced by factors such as BGP configuration errors, OSPF neighbor adjacency issues, or ISIS LSP database inconsistencies. Furthermore, the Arista EOS implementation of these protocols may exacerbate the situation due to its unique feature set and configuration nuances. The combination of these factors may lead to a cascading effect, culminating in a critical failure state, where the access node BAS-ACC-62 becomes non-functional, resulting in severe service disruptions.

Pre-requisite Checks

- check: Verify that the Basavanagudi POP is accessible and that remote hands are on-site to perform any necessary physical interventions.

- check: Check the status of the redundant link at the Basavanagudi POP to ensure that it is operational and ready to assume the traffic load in case of a failure.

- check: Verify that all power supplies are functioning correctly and that there are no signs of overheating or other environmental issues affecting the node.

Potential Root Causes

Cause	Probability
BGP configuration errors	High
OSPF neighbor adjacency issues	Medium
ISIS LSP database inconsistencies	Low

Diagnostic Commands & Expected Outputs

```
command: show bgp summary, expected_output: BGP table version is 1, local AS number 65000, description: This command will provide an overview of the BGP routing table, including the number of prefixes and neighbors. It will help identify any BGP configuration errors or adjacency issues.
```

```
command: show ospf neighbors, expected_output: Neighbor ID Pri State
Dead Time Address Interface, description: This command will display
the OSPF neighbor adjacency status, including the state and dead time.
It will help identify any OSPF neighbor adjacency issues.

command: show isis database, expected_output: ISIS LSP database
(sequence 0x4) LSP ID: 100.100.100.100, description: This command will
display the ISIS LSP database, including the LSP ID and sequence
number. It will help identify any ISIS LSP database inconsistencies.
```

Resolution Steps

Step 1: step: Verify and correct BGP configuration errors, command: show run | include bgp, description: Verify the BGP configuration and correct any errors. This may involve checking the BGP AS number, neighbor IP addresses, and timer settings.

Step 2: step: Reset OSPF neighbor adjacency, command: clear ospf neighbor, description: Reset the OSPF neighbor adjacency to re-establish a stable connection with the neighbor routers.

Step 3: step: Flush ISIS LSP database, command: clear isis lsp, description: Flush the ISIS LSP database to remove any inconsistent LSPs and re-establish a stable connection with the neighbor routers.

Post-Check Verification

```
command: show bgp summary, expected_output: BGP table version is 1,
local AS number 65000, description: Verify that the BGP routing table
is stable and that there are no errors.

command: show ospf neighbors, expected_output: Neighbor ID Pri State
Dead Time Address Interface, description: Verify that the OSPF
neighbor adjacency is stable and that there are no errors.

command: show isis database, expected_output: ISIS LSP database
(sequence 0x4) LSP ID: 100.100.100.100, description: Verify that the
ISIS LSP database is stable and that there are no errors.
```

Roll-back Procedures

- step: Re-apply the previous configuration, command: copy running-config startup-config, description: Re-apply the previous configuration to roll back the changes.

- step: Clear the ISIS LSP database, command: clear isis lsp, description: Clear the ISIS LSP database to remove any inconsistent LSPs.

- step: Reset OSPF neighbor adjacency, command: clear ospf neighbor, description: Reset the OSPF neighbor adjacency to re-establish a stable connection with the neighbor routers.